

Logistic Regression

Linear Regression

- Given x , we approximate y with a linear function.

$$h(x) = \sum_{i=0}^d \theta_i x_i = \theta^T x$$

- θ is called parameters/weights

- What about classification?

Classification

- Y is a discrete target space

- Commonly binary classification tasks

Yes/No predict

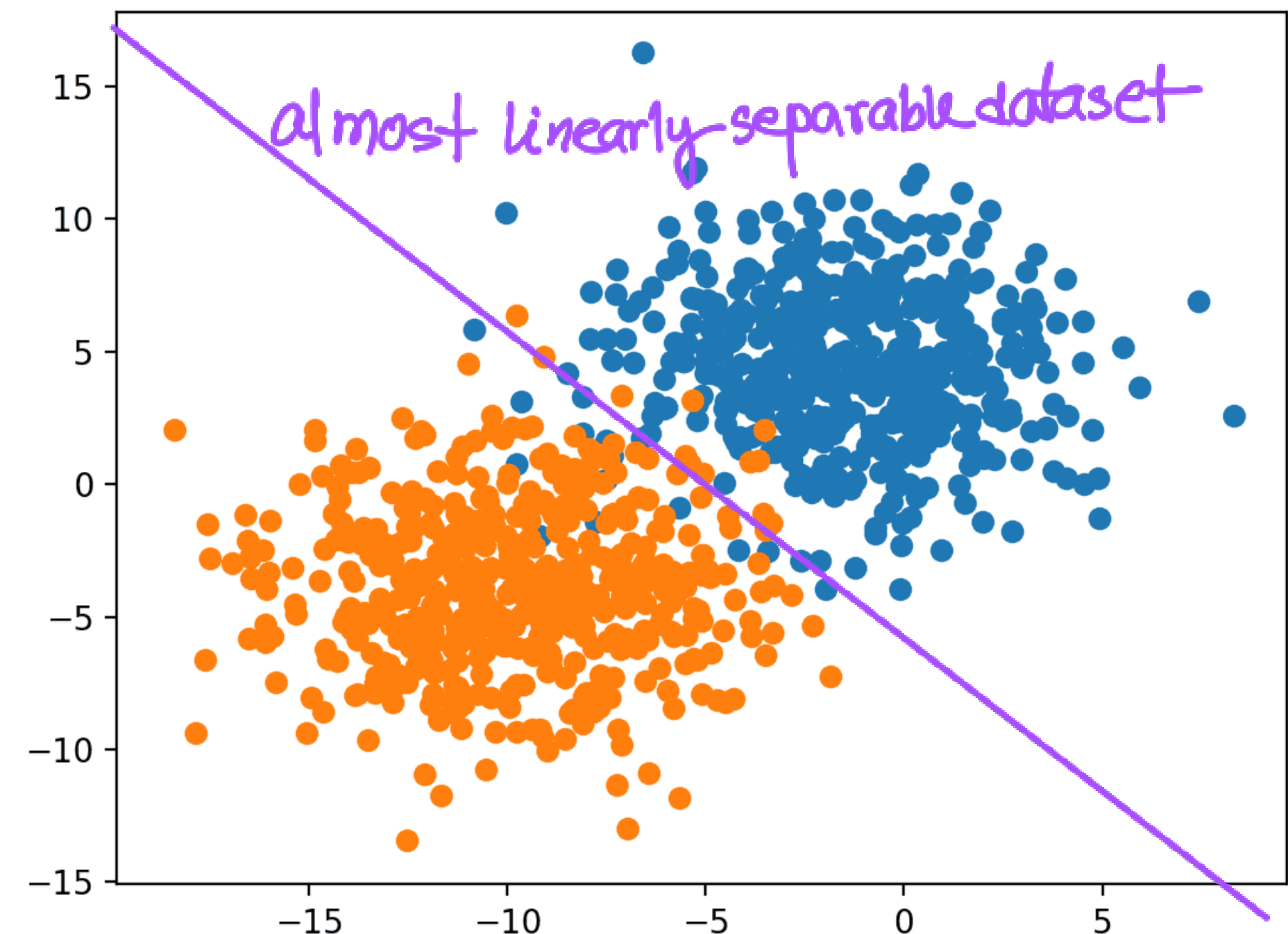
- $y \in \{0, 1\}$ ✓

০-১, +১/-১ ডকুমেন্ট

- Often called positive class vs negative class

+1

০ কে (-১) or negative
চিহ্ন করা হয়



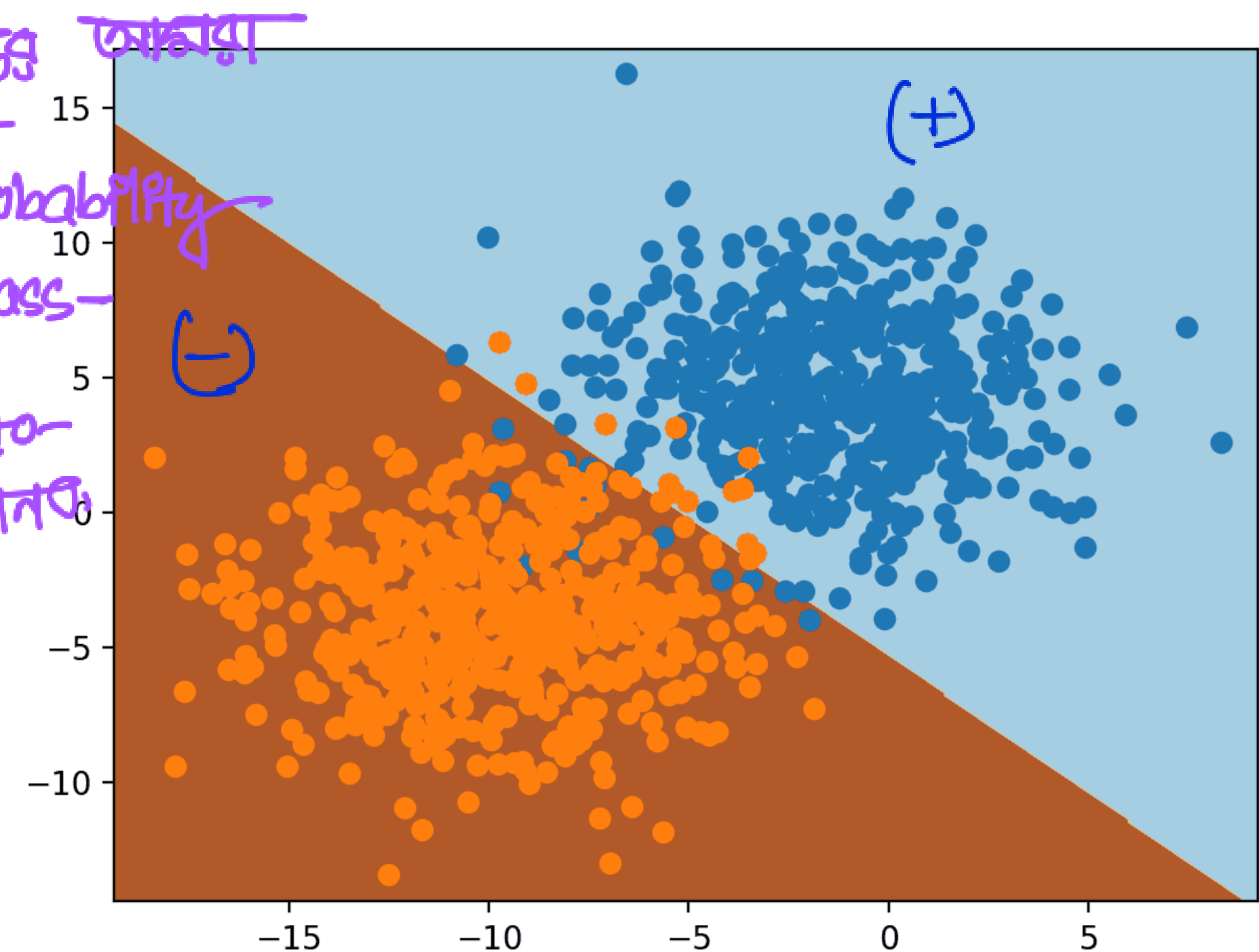
Logistic Regression

- Instead of predicting $y \in \{0, 1\}$, we can predict $\mathbb{P}(y = 1)$

$$h_{\theta}(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$

$$g(z) = \frac{1}{1 + e^{-z}}$$

Why this function?



Classification with Linear Classifier

		Range		
	To Predict	Min	Middle	Max
Binary Classification	y	0	-	1
Predict Probability	$p = P(y = 1), q = 1 - p$ <i>→ এর range 0 ≤ p ≤ 1</i>	0 <i>→ এর ২য় স্তর probability 0.5 0.5</i>	0.5 <i>middle ground of p</i>	1 <i>→ এর ২য় স্তর probability 0.5 1</i>
<u>Odds</u> → <i>→ ২ class এর সম্ভাব্য prob</i>	$\frac{p}{1 - p}$	0	1	∞
<u>Log-odds</u> <i>So we can use log-odds</i>	$\log\left(\frac{p}{1 - p}\right)$	$-\infty$	0	∞
Linear Classifier Prediction <i>Regression</i>		$-\infty$	0	∞

→ এখানে থেকে value $-\infty$ to $+\infty$ এর মধ্যে থাকতে পারে।

Logistic Regression

$$h_{\theta}(x) = \theta^T x = \log \left(\frac{p}{1-p} \right)$$

$$e^{\theta^T x} = \frac{p}{1-p}$$

$$e^{\theta^T x} - e^{\theta^T x} p = p$$

$$e^{\theta^T x} = p(1 + e^{\theta^T x})$$

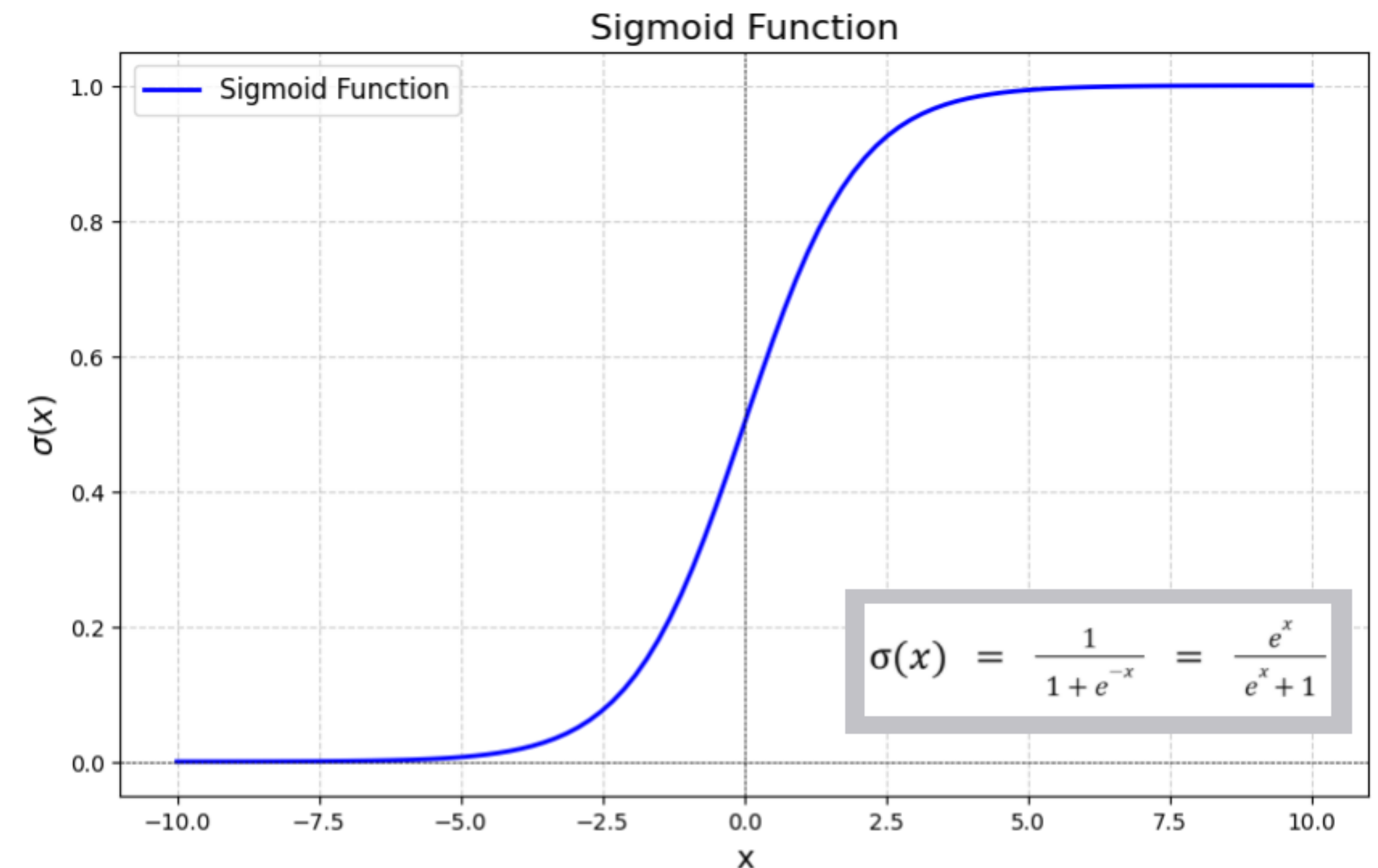
$$p = \frac{e^{\theta^T x}}{1 + e^{\theta^T x}}$$

$$p = \frac{1}{1 + e^{-\theta^T x}}$$

$$P(y = 1) = p = \sigma(\theta^T x)$$

(sigmoid function)

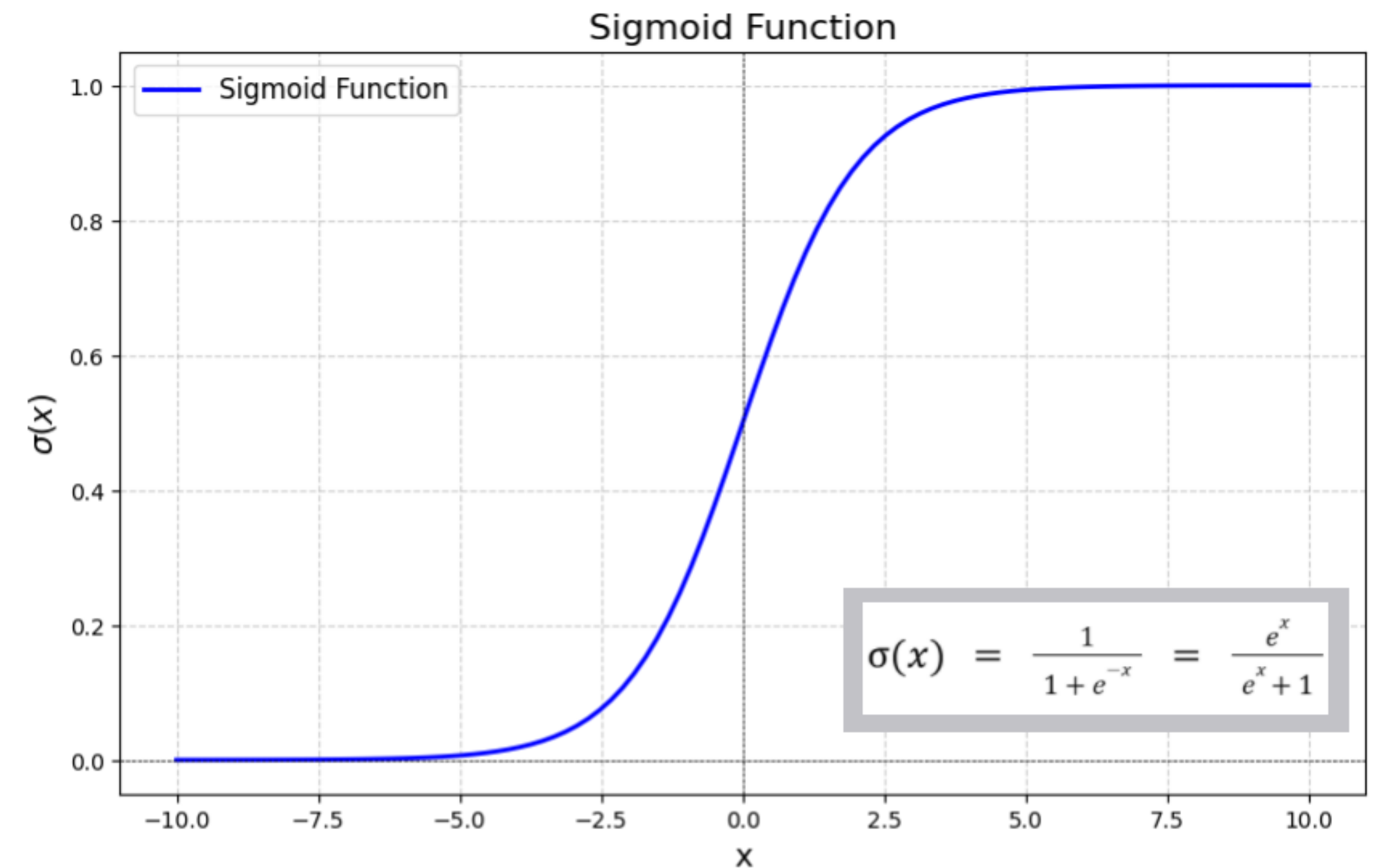
→ one exact best & unique way to get best classifier.



Sigmoid Function

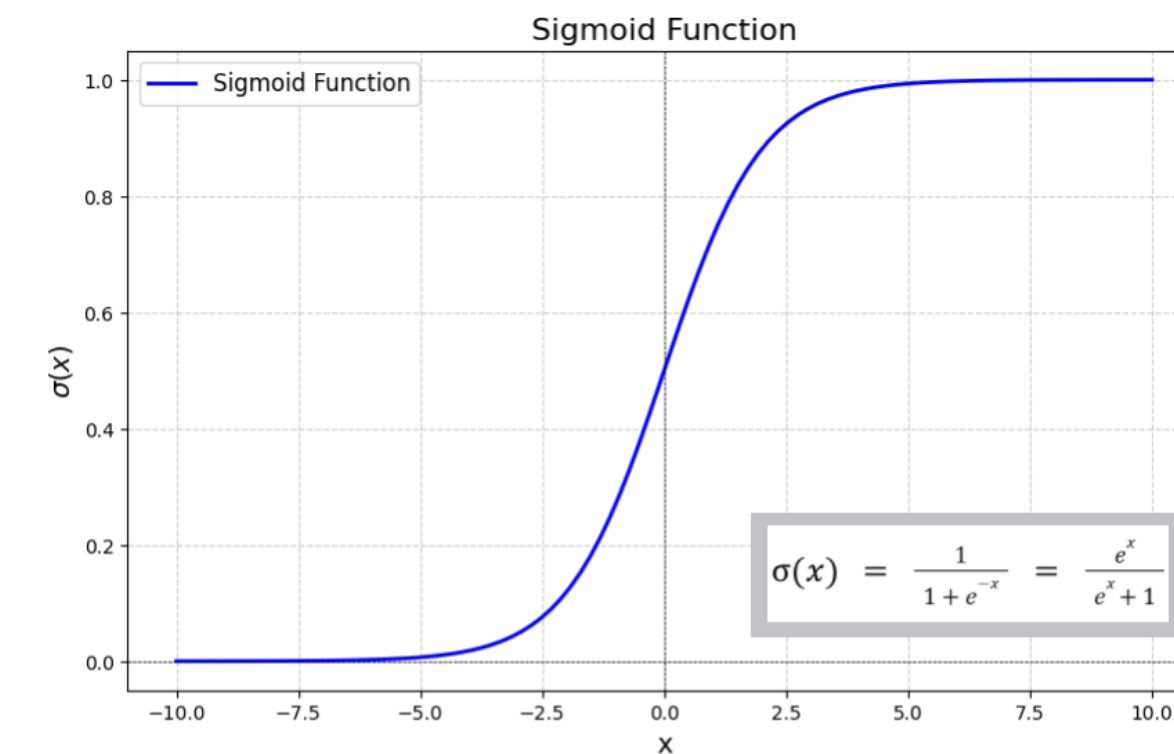
- Some properties,

$$\checkmark g'(z) = g(z)(1 - g(z)).$$



Logistic Regression

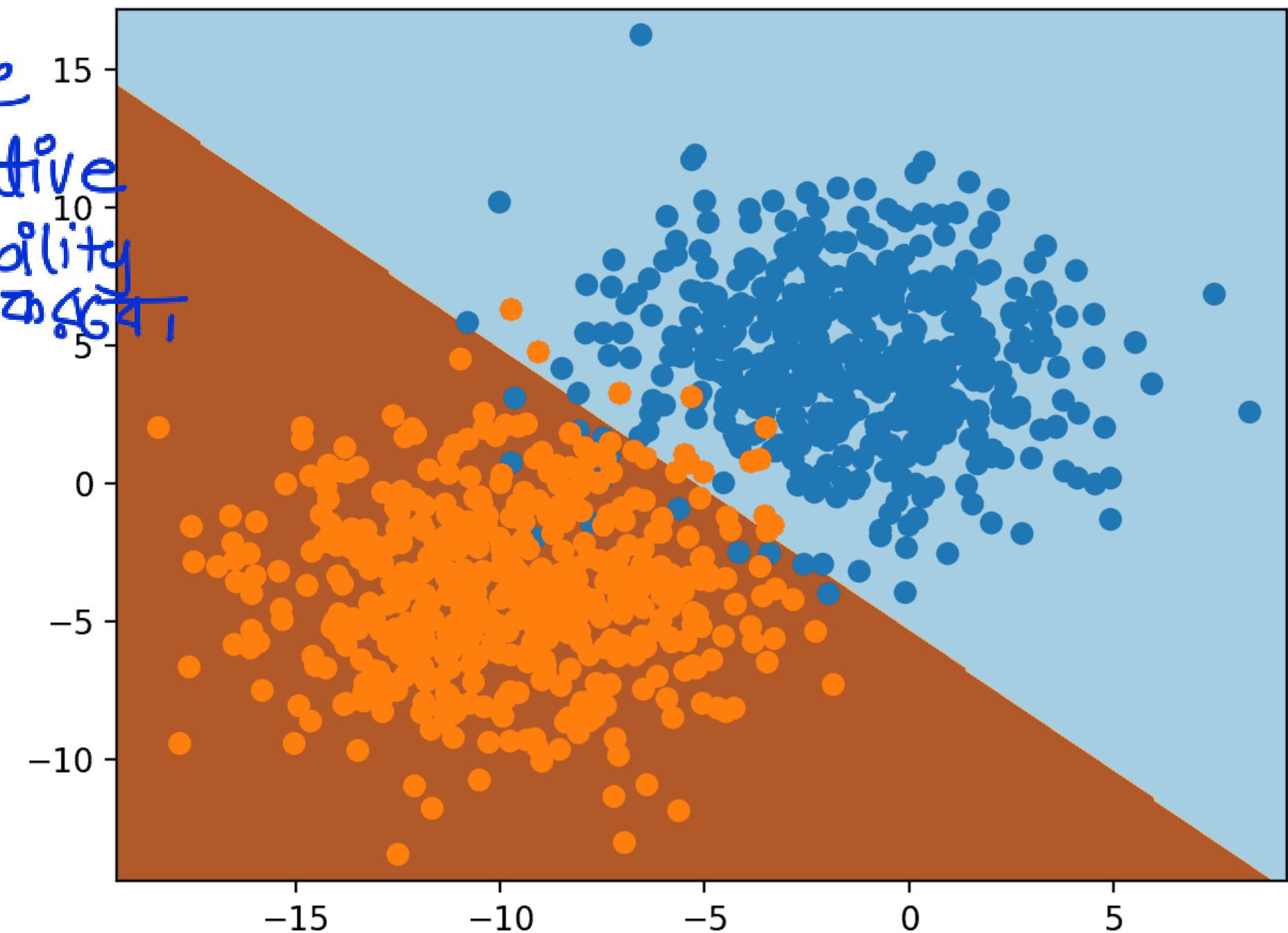
$$h_{\theta}(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}},$$



- How to choose θ ?

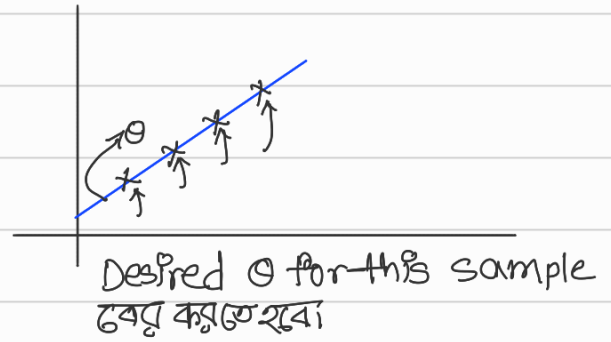
→ positive sample
Higher and negative
sample Lower probability
predict કરશે,
exactly value predict ના કરાશે,
probability predict કરશે.

- Maximum Likelihood Estimation



(x, y) dataset দেওয়া আছে আমাদের বের করতে হবে কোন θ টি Desired.

$$\frac{P(\theta|D)}{\rightarrow \text{maximize } \theta}$$



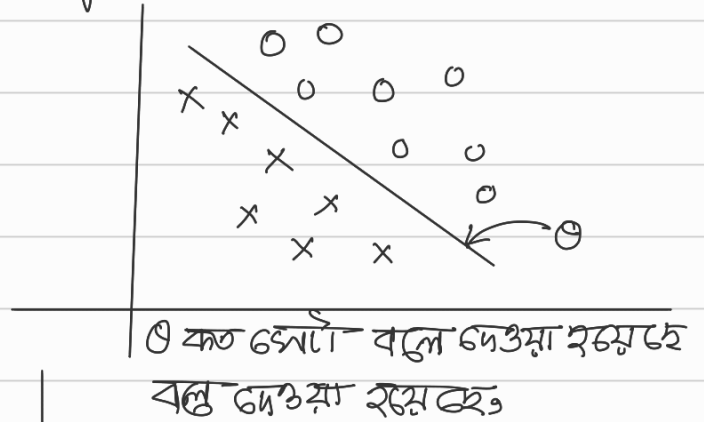
Posterior $\propto$ Likelihood $\times$ Prior

θ এর উপর আমরা কোনো prior knowledge নাই।

$$\begin{aligned} P(D|\theta) \\ &= P(x, y|\theta) \\ &= P(y|x;\theta) P(x|\theta) \end{aligned}$$

(dataset is the ground truth দিয়ে দেওয়া হয়েছে, it is not the prediction)

Finally, আমরা জানি $P(y|x;\theta)$ term টিকে predict করি। তাই এটাকে maximize করতে we can meet our target.



$$y = f(x;\theta)$$

আমাদের Desired task likelihood কে estimate & maximize করা।

$$P(D|\theta) \leftarrow$$

Probability (Recap)

Bayes' Theorem

- Given two events A and B,

$$P(A, B) = P(A|B)P(B) = P(B|A)P(A) \checkmark$$

and,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \checkmark$$

Probability (Recap)

Bayes' Theorem

- Given (training) data D and a parameter θ (*random variable*),
→ θ learn করতে হবে
- Likelihood, $P(D|\theta)$
- Prior, $P(\theta)$
- Evidence, $P(D)$ *→ Dataset appear করতে probability কত?*
- Posterior,

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

→ fixed → which is actually $P(x,y)$

Posterior $\propto$ Likelihood $\times$ Prior

→ try predict করতে try করছি।
 $P(y=1|x;\theta)$ = $P(x,y|\theta)$ = $P(y|x;\theta)P(x|\theta)$
probability of $y=1$ করতে, given x input, θ parameter
→ এখন maximize করতে, we can do our classification task.

Logistic Regression

- We assume

$$P(y = 1 \mid x; \theta) = h_{\theta}(x)$$

$$P(y = 0 \mid x; \theta) = 1 - h_{\theta}(x)$$

Or,

$$p(y \mid x; \theta) = (h_{\theta}(x))^y (1 - h_{\theta}(x))^{1-y}$$

- Likelihood

$$L(\theta) = p(\vec{y} \mid X; \theta)$$

$$= \prod_{i=1}^n p(y^{(i)} \mid x^{(i)}; \theta)$$

$$= \left(\prod_{i=1}^n \right) (h_{\theta}(x^{(i)}))^{y^{(i)}} (1 - h_{\theta}(x^{(i)}))^{1-y^{(i)}}$$

→ ଏହାକୁ sample likelihood
କୁହାଯାଏ।

ଏହି product ଠାରୁ remove ଏବଂ କାର୍ଯ୍ୟ we will
use log.

Logistic Regression

- Likelihood

$$\begin{aligned} L(\theta) &= p(\vec{y} \mid X; \theta) \\ &= \prod_{i=1}^n p(y^{(i)} \mid x^{(i)}; \theta) \\ &= \prod_{i=1}^n (h_{\theta}(x^{(i)}))^{y^{(i)}} (1 - h_{\theta}(x^{(i)}))^{1-y^{(i)}} \end{aligned}$$

- Log Likelihood

$$\begin{aligned} \ell(\theta) &= \log L(\theta) \\ &= \sum_{i=1}^n \underline{y^{(i)}} \log h(x^{(i)}) + \underline{(1 - y^{(i)})} \log(1 - h(x^{(i)})) \end{aligned}$$

analytic form (যাও ডিফারেন্সিয়েট করা
কঠিন hard) (difficult?)

So we use gradient ascent.

Logistic Regression

$$\sigma'(x) = \sigma(x) (1 - \sigma(x))$$

$$h_{\theta}(x) = \sigma(\theta^T x) = \sigma(\theta^T x) (1 - \sigma(\theta^T x))$$

- Maximize log likelihood with,
- Stochastic Gradient Ascent

$$\theta_j := \theta_j + \alpha (y^{(i)} - h_{\theta}(x^{(i)})) x_j^{(i)}$$

- Gradient Ascent

$$\theta_j := \theta_j + \alpha \sum_{i=1}^m (y^{(i)} - h_{\theta}(x^{(i)})) x_j^{(i)}$$

Maximum Likelihood estimation is basically $\sum_{i=1}^m$ parameter estimation कर रहे।

Perceptron

- We can modify the logistic regression method to “force” it to output values that are either 0 or 1 or exactly.

$$h_{\theta}(x) = \underline{g(\theta^T x)}$$

কম্পিউটার sigmoid না assume
করবে $g(z)$ এর মাধ্যমে হ্যান্ড prediction করা হবে

- Where

$$g(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

- With the same update rule,

$$\theta_j := \theta_j + \alpha (y^{(i)} - h_{\theta}(x^{(i)})) x_j^{(i)}$$

Reference

- CS229 Lecture Note
- Section 2.1